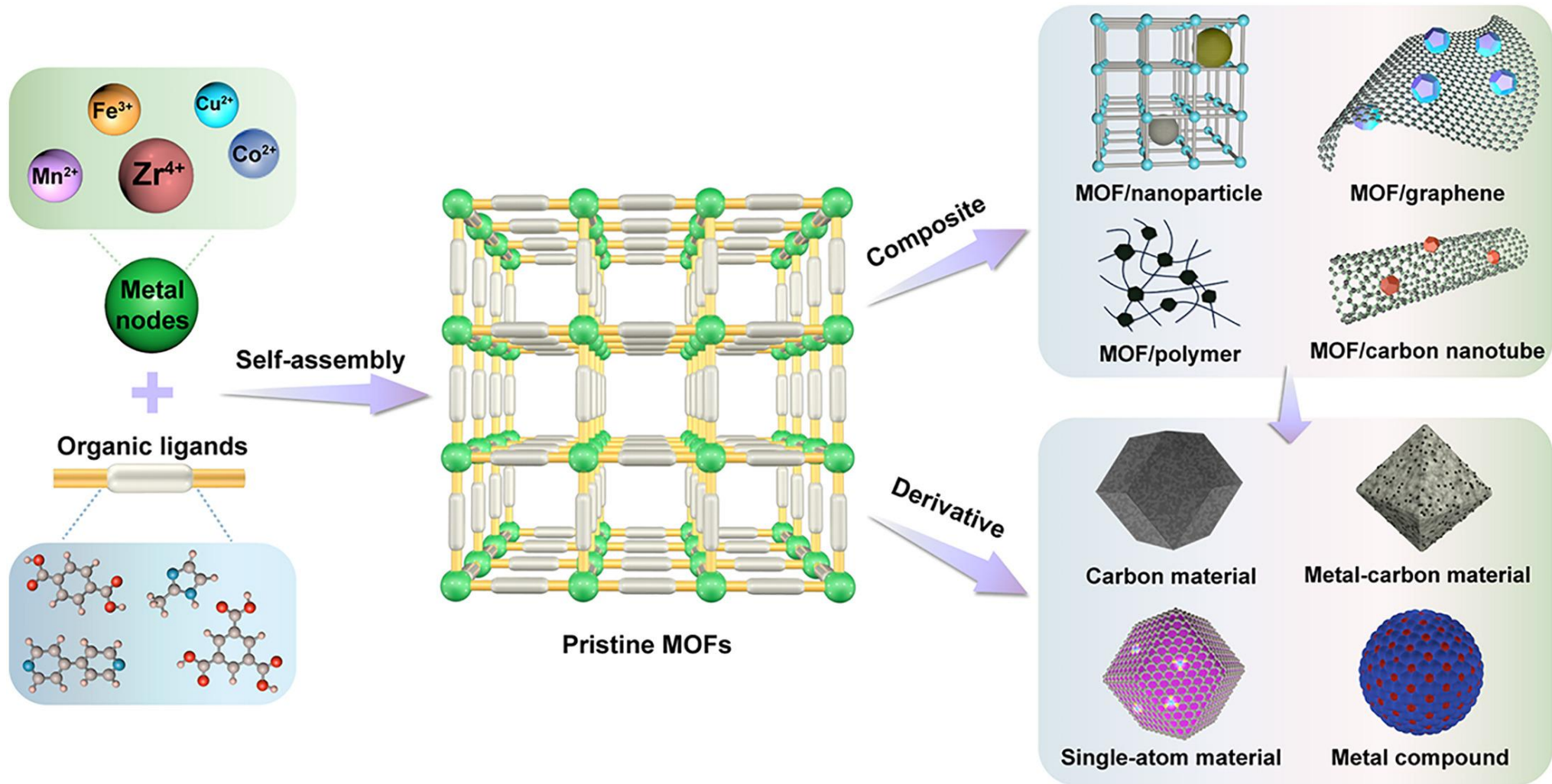
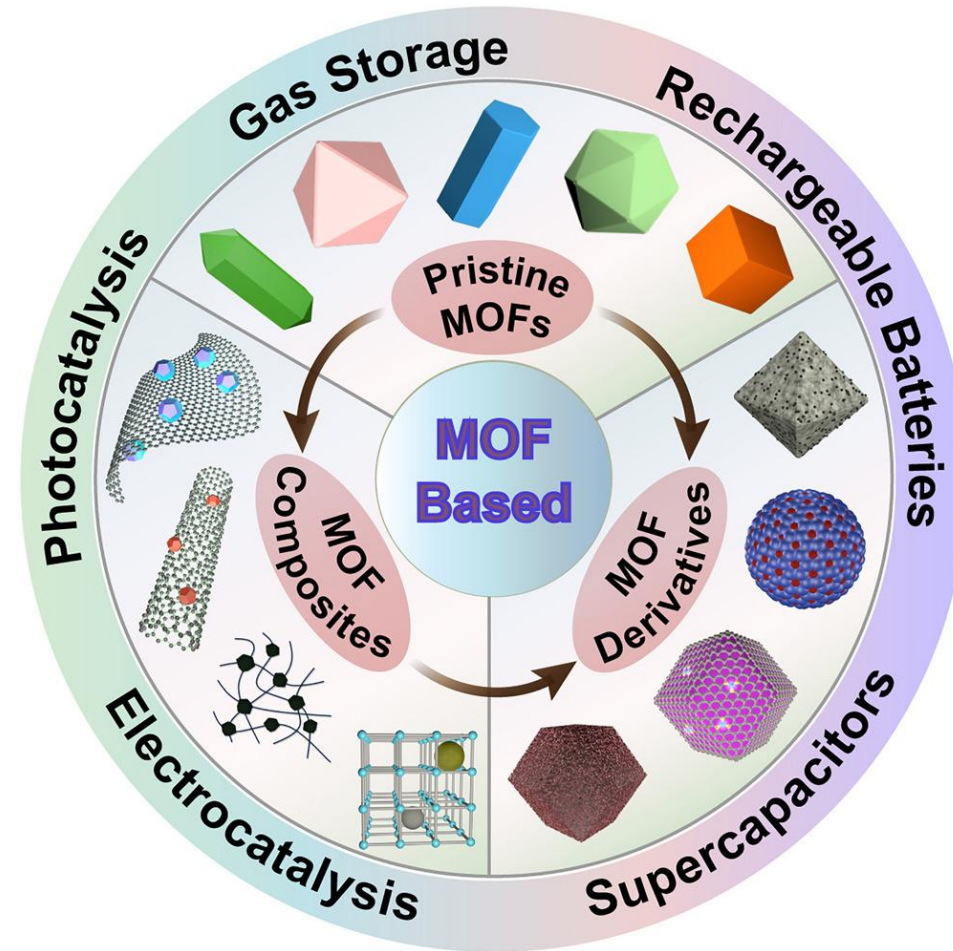


MOF's Applications

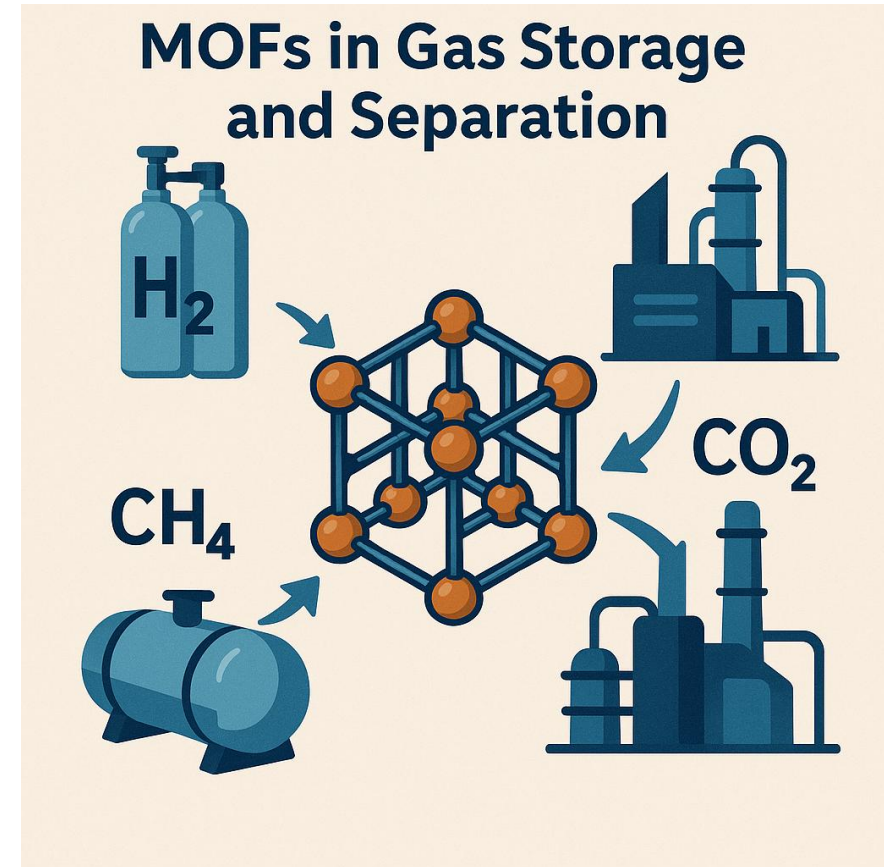
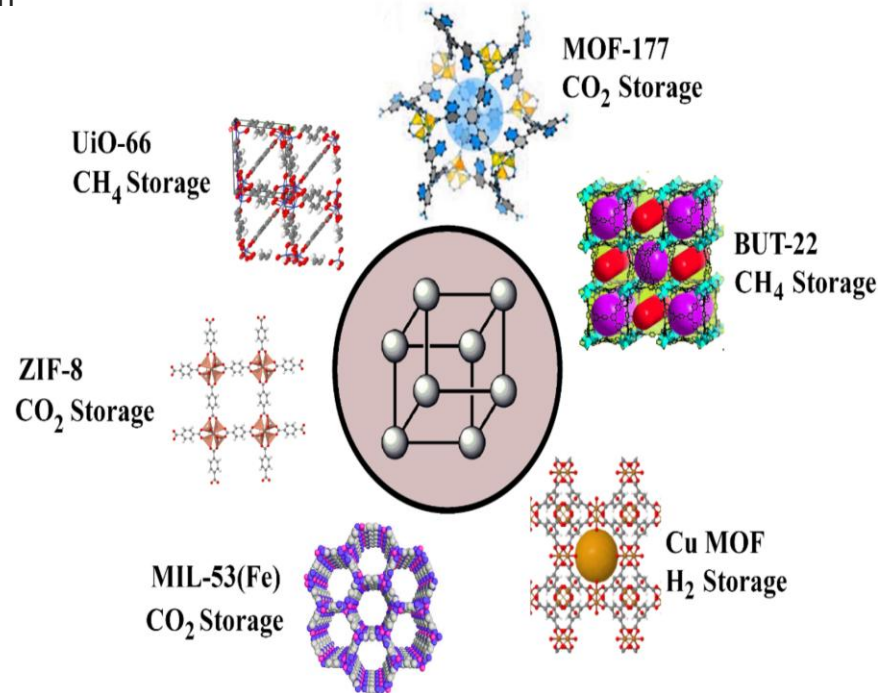


MOF's Applications



MOF's Applications : Gas storage and Separation

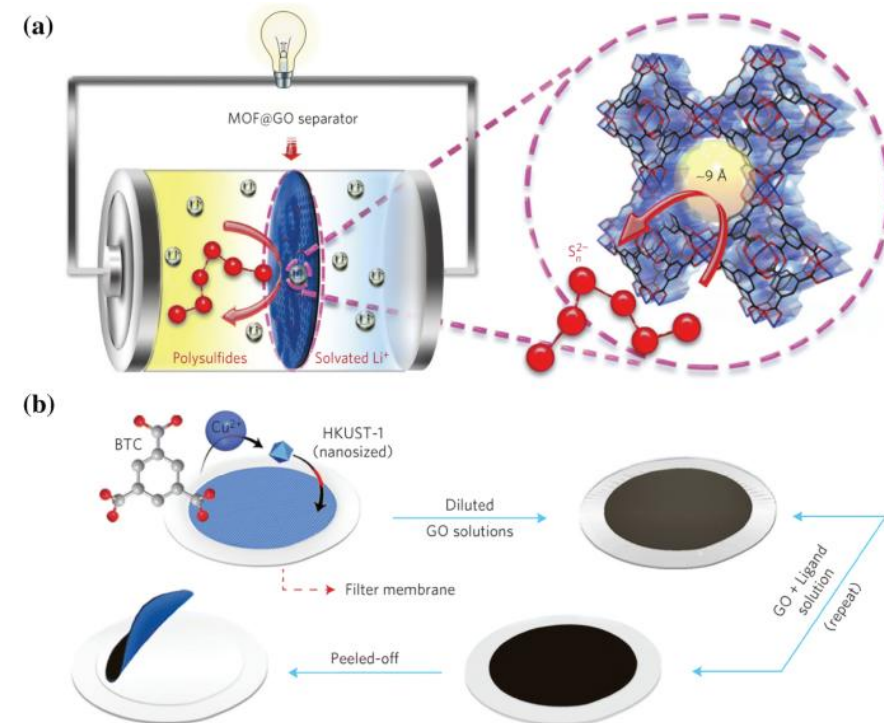
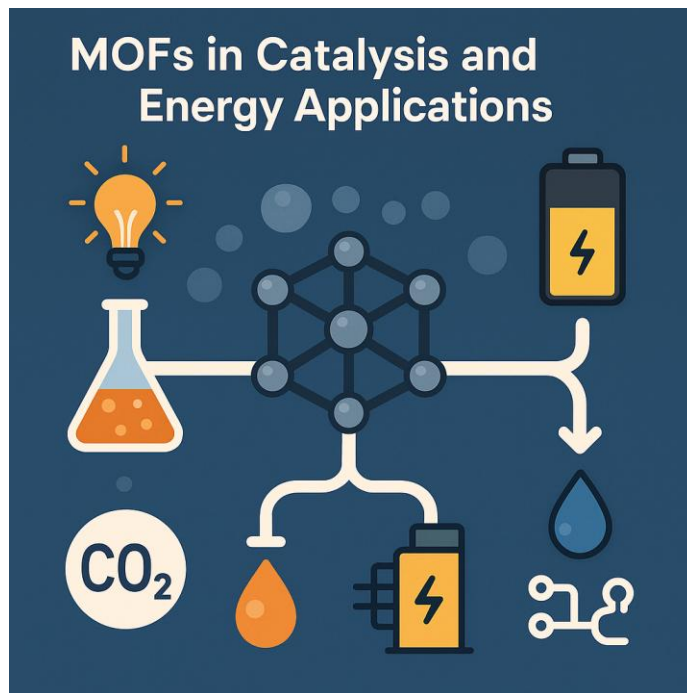
MOFs high porosity and tunable pore environments provide significant advantages over traditional adsorbents. Their internal surface areas can reach over 7,000 m²/g, making them highly effective for capturing and storing various gases — from energy carriers like hydrogen (H₂) and methane (CH₄) to greenhouse gases like carbon dioxide (CO₂).



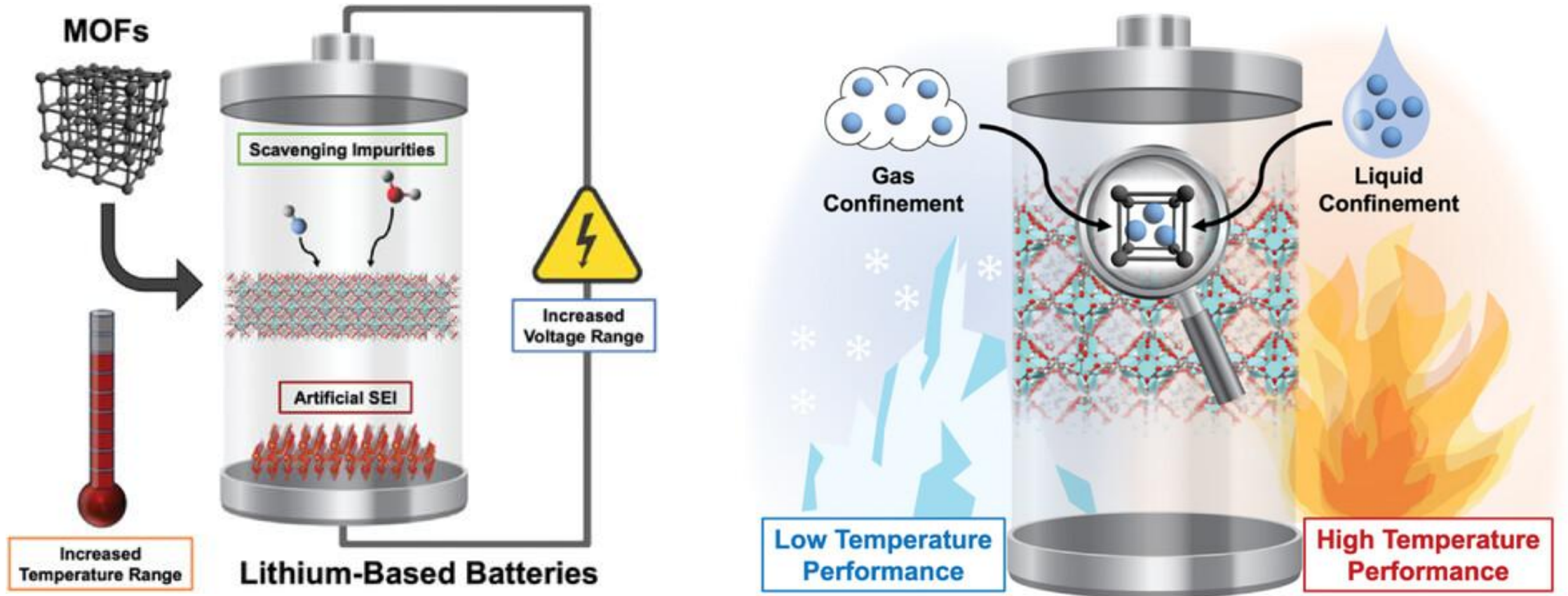
Different MOFs for gas storage applications.

MOF's Applications: Electrochemical Application

MOFs energy-conversion reactions owing to the distinct advantages of structural diversity, porous structures, and tailorable catalytic sites. They are highlighted as appropriate electrocatalysts with high activity and stability for various frontier reactions, including oxygen evolution reaction (OER), hydrogen evolution reaction (HER), oxygen reduction reaction (ORR), and carbon dioxide reduction reaction (CO₂RR).

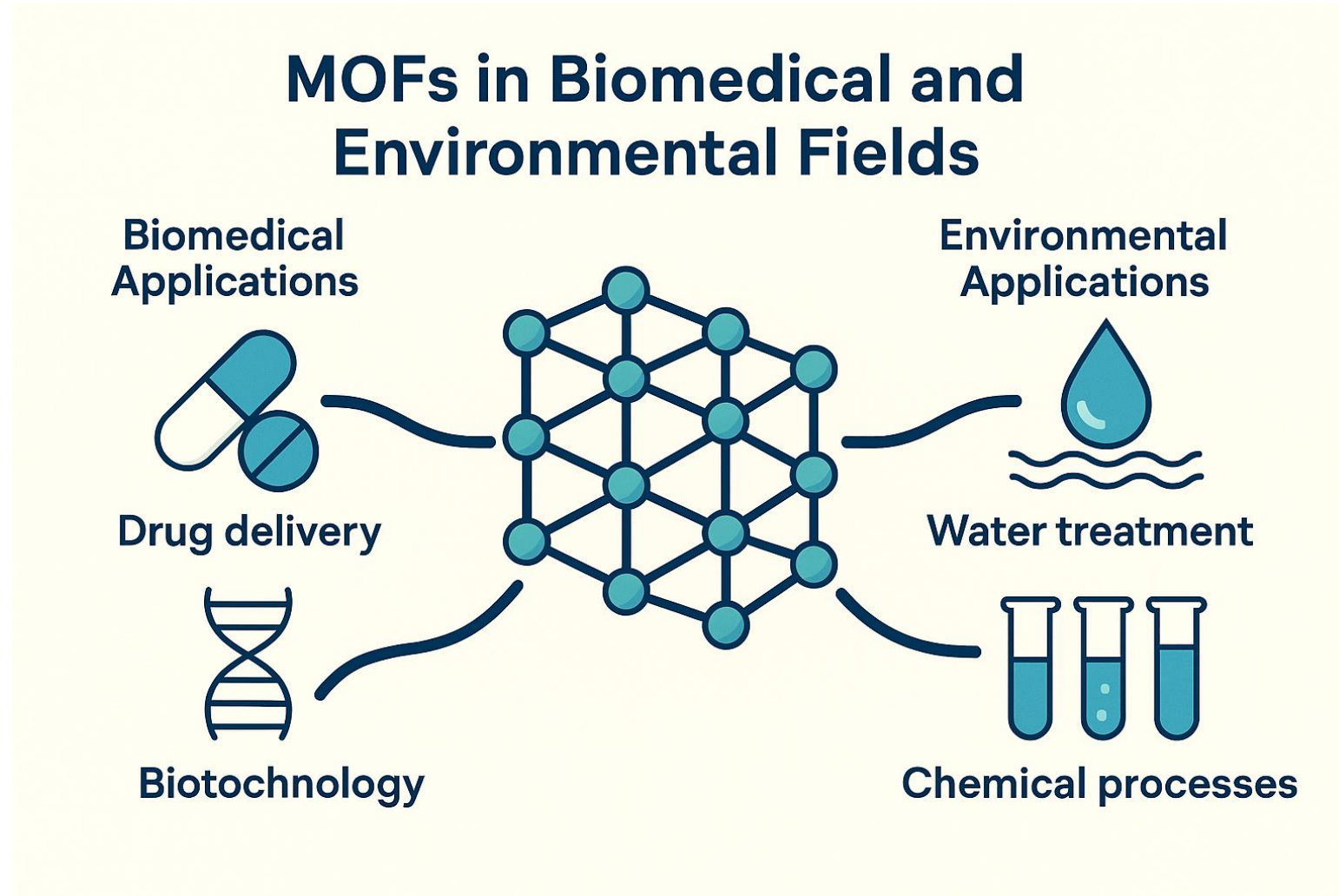


MOF's Applications: Electrochemical Application



Schematic showing the MOF-based electrolytes for batteries operating under extreme temperatures.

MOF's Applications: Biomedical and Environmental Fields



MOF Membrane for Wastewater Treatment

Membrane technology

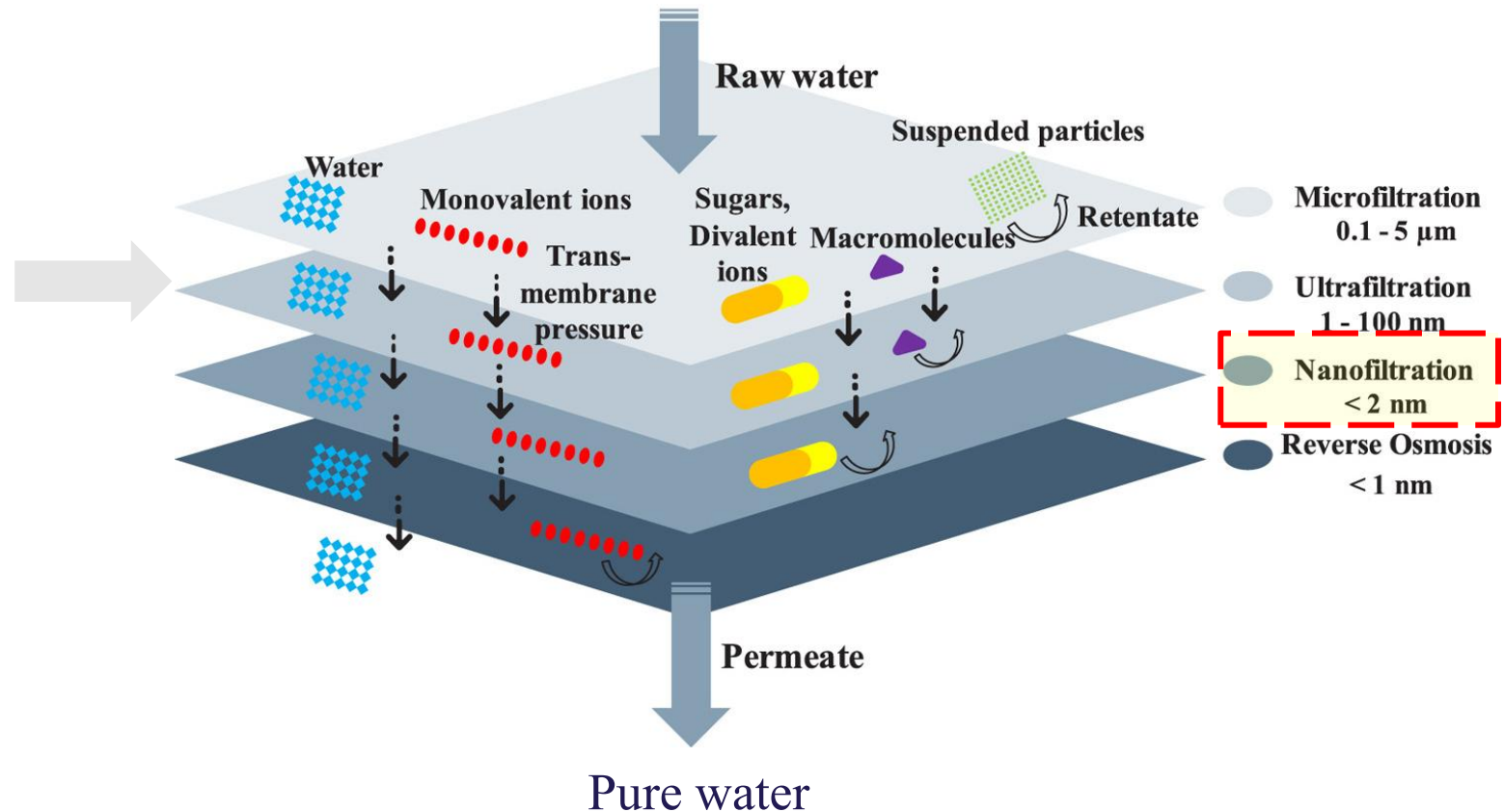
Pressure

Higher
contaminant
concentration

Semipermeable
membrane

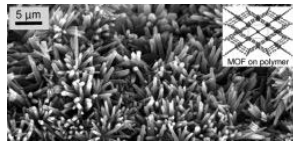
Lower
contaminant
concentration

Membrane filtration Classification



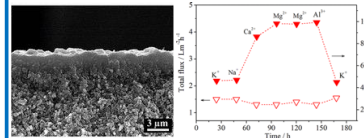
MOF Membrane for Wastewater Treatment

First MOF membranes prepared by microwave on polymer substrate



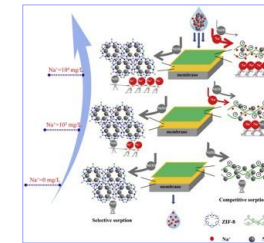
J. Am. Chem. Soc., 2010, 132:15687-15691

UIO-66 fibers for desalination



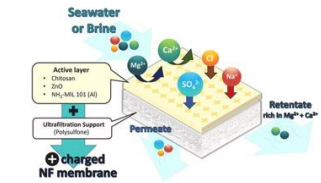
I. Am. Chem. Soc. 2015, 137(22), 6999-7002.

MOF hybrid membrane firstly used for removing trace Ni(II)



Water Res., 2018, 143:87-98

NH2-MIL-101(Al) nanofiltration membrane



J. Membr. 2023, 685, 121941

2005

2010

2012

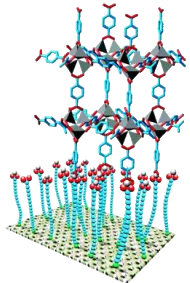
2015

2018

2021

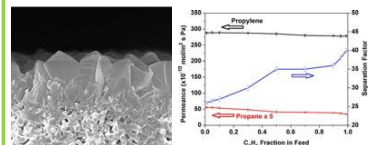
2023

First bare MOF (MOF-5 membranes)



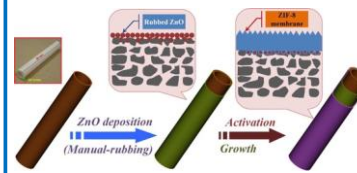
J. Am. Chem. Soc. 2005, 127, 40.13744-13745.

ZIF8 membrane with long-term stability for separation



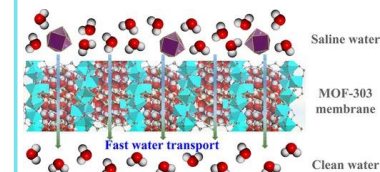
J. Membr. Sci. 2009, 339(1-3), 93-98

Mechanical stability ZIF-8 membrane for separation



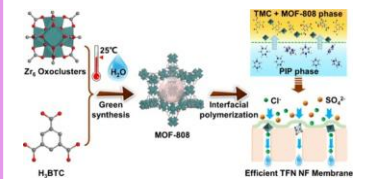
J. Membr. Sci. 2015, 490, 354-363

MOF-303 Membranes for Desalination



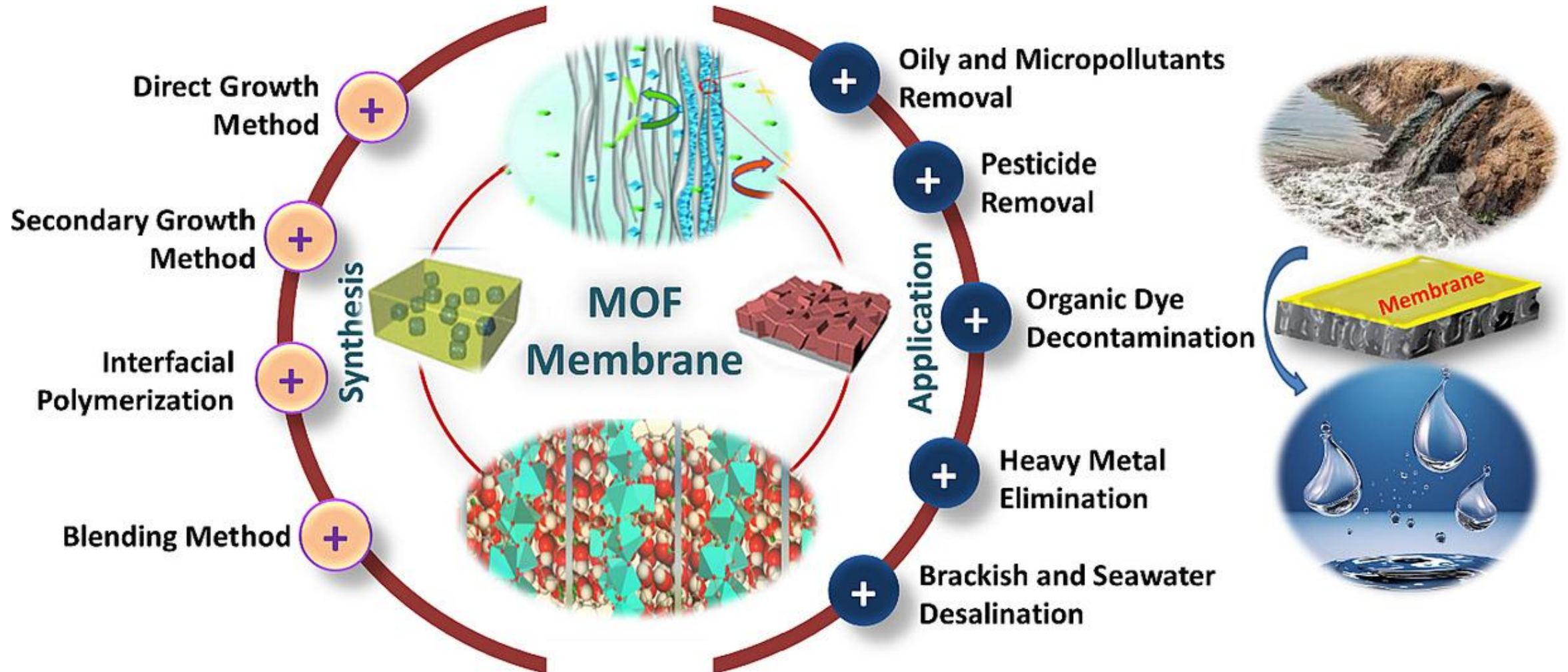
J. Am. Chem. Soc. 2021, 143 (48), 20055-20058

MOF-808/Polyamide Nanofiltration Membranes

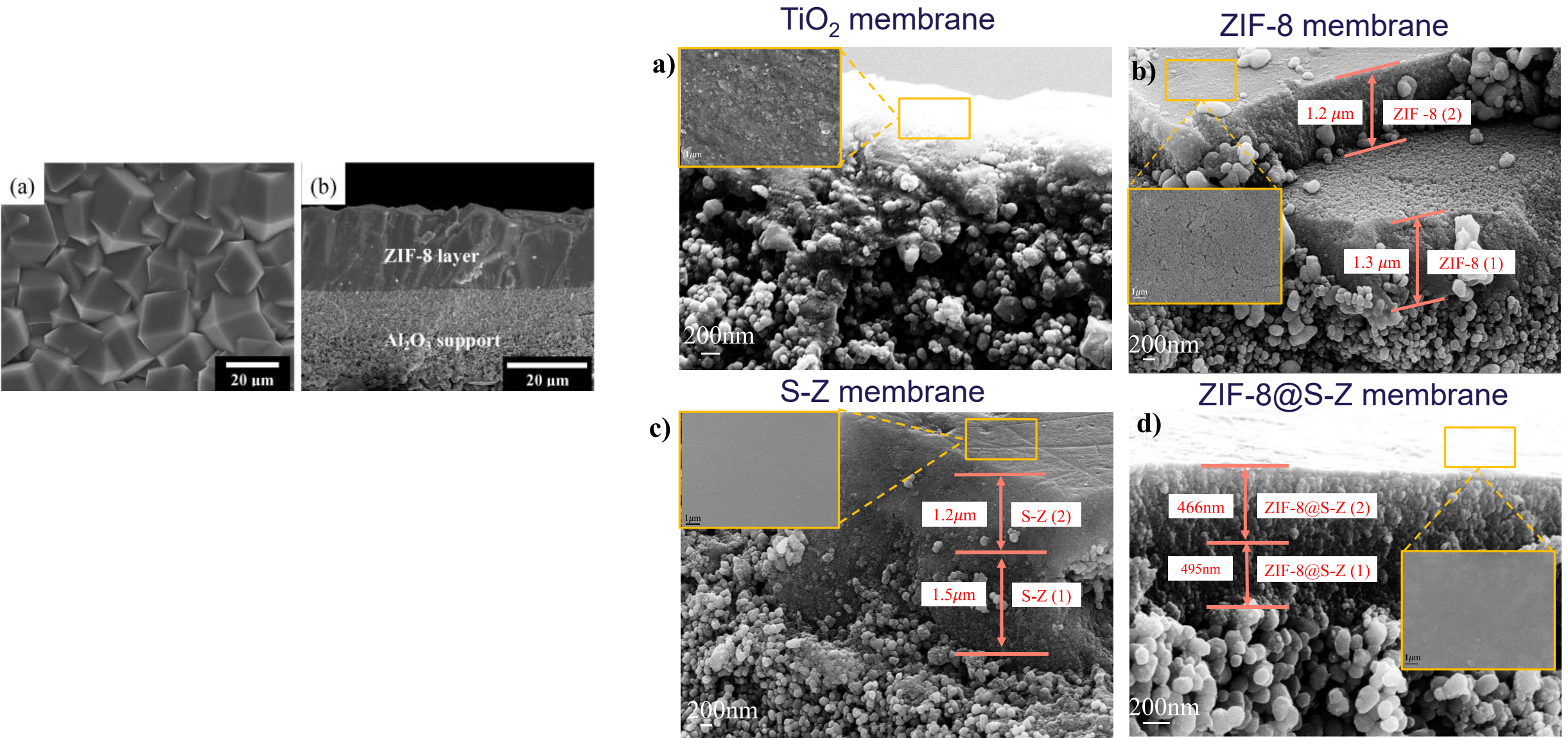


ACS Appl. Nano Mater. 2023, 6 (19), 17615-17625

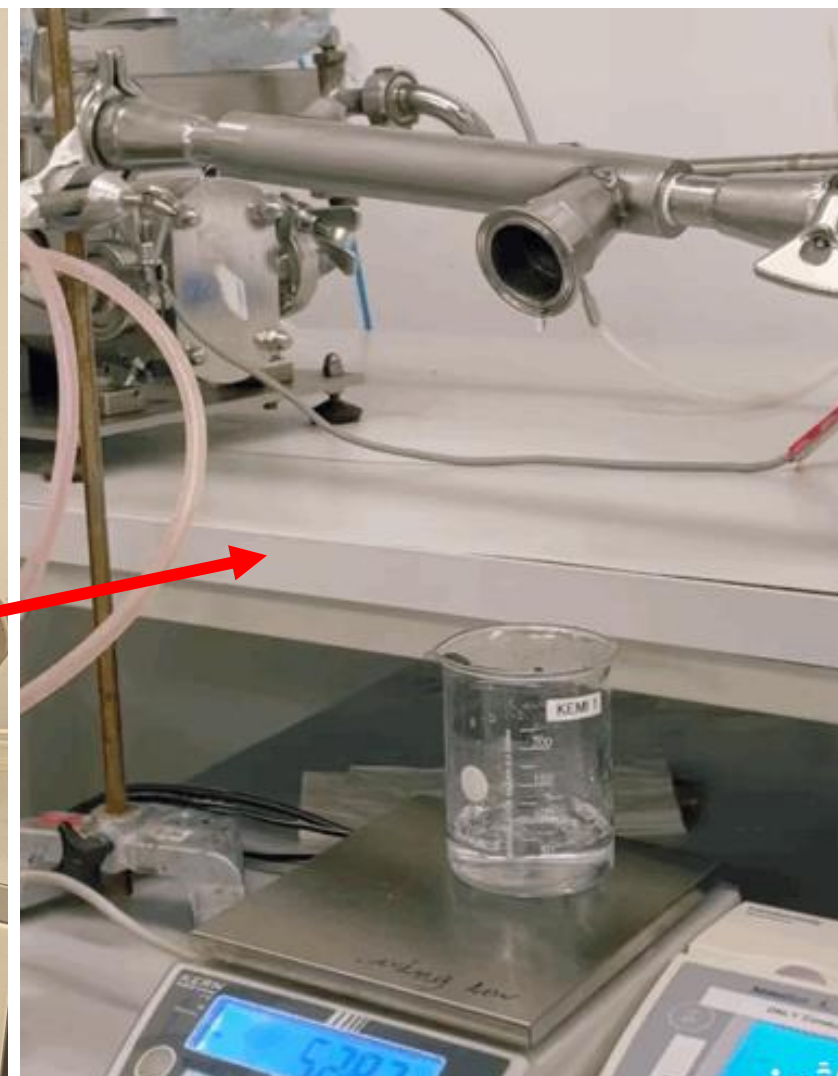
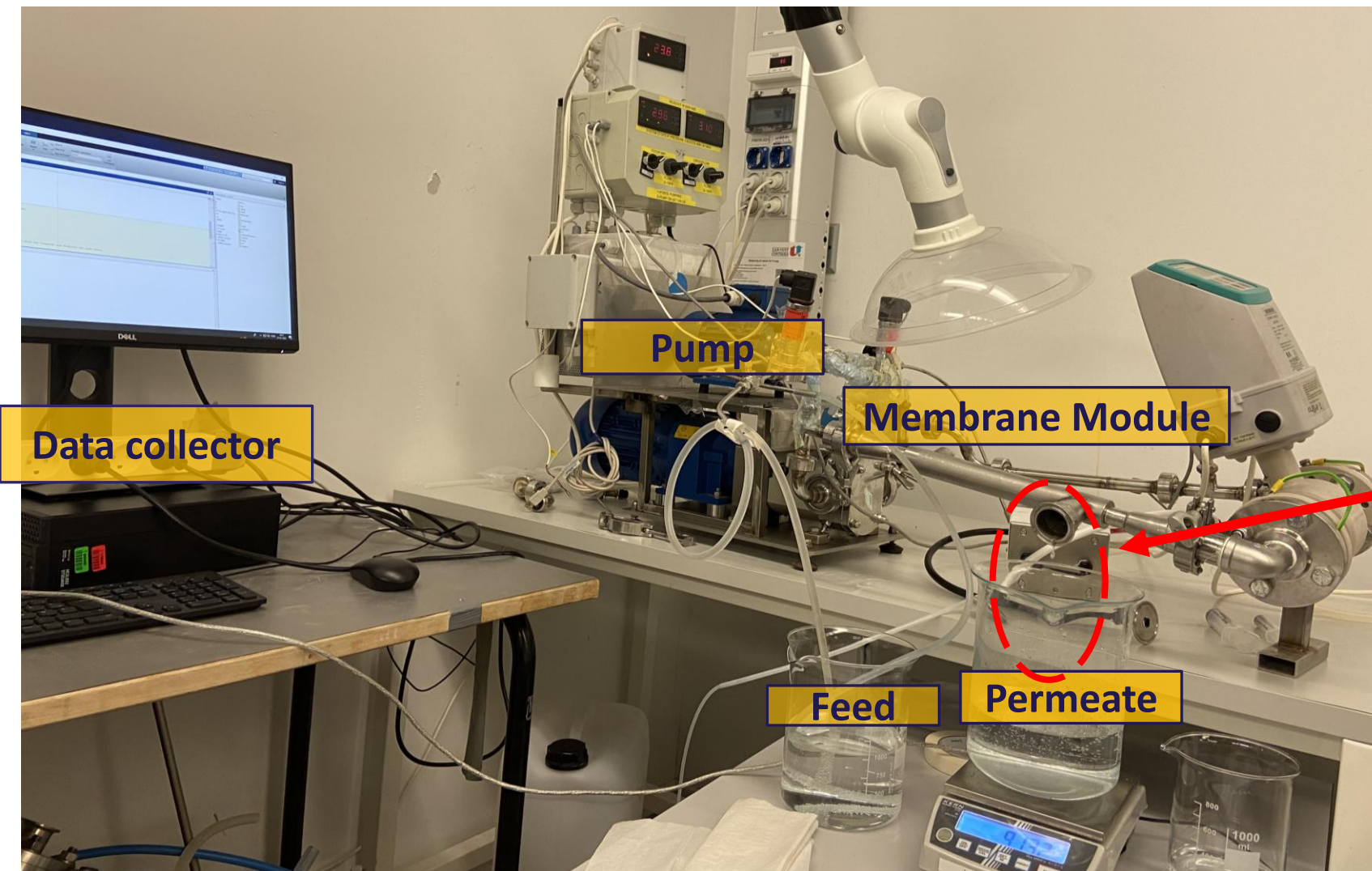
MOF Membrane for Wastewater Treatment



MOF Membrane for Wastewater Treatment

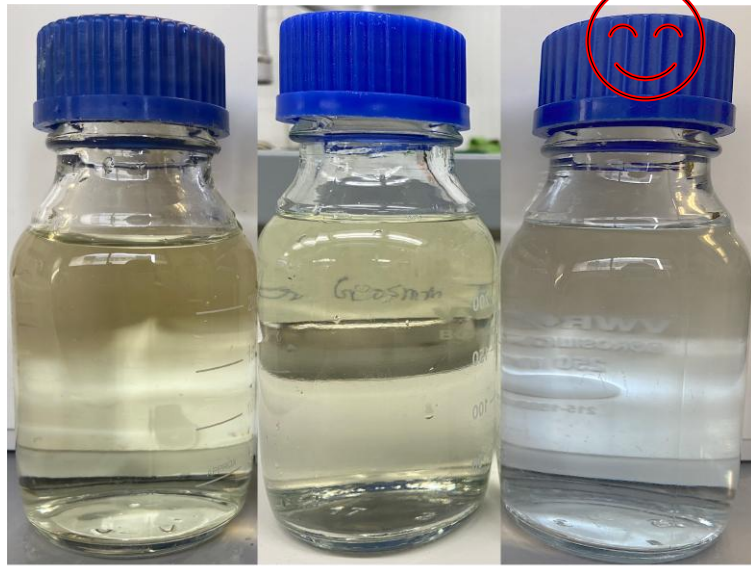


MOF Membrane for Wastewater Treatment



MOF Membrane for Wastewater Treatment

Danish Fish Farm Samples



Aquaculture
effluent (feed)

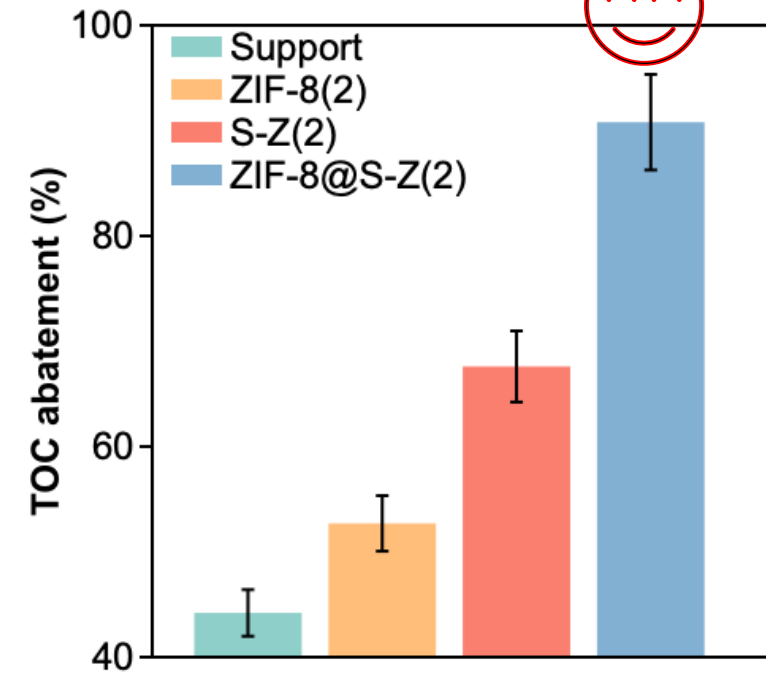
Support
permeate

ZIF-8@S-Z-2
permeate

yellowish
turbid

Colorless
clear

TOC abatement



MOF Membrane for Wastewater Treatment

